

Lucas C. Wheeler, PhD

POSTDOCTORAL ASSOCIATE · DEPARTMENT OF ECOLOGY AND EVOLUTIONARY BIOLOGY · CU BOULDER

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Education

Ph.D. in Biochemistry

UNIVERSITY OF OREGON

Eugene, OR

Sep. 2012 - Dec. 2017

- Department: Chemistry and Biochemistry
- Advisor: Michael Harms
- Dissertation: Evolution of Metal and Peptide Binding in the S100 Protein Family.

B.S. in Biochemistry

MONTANA STATE UNIVERSITY

Bozeman, MT

Sep. 2009 - June 2012

- Department: Chemistry and Biochemistry
- Advisor: Trevor Douglas

Work Experience

Postdoctoral Fellow

UNIVERSITY OF SOUTH CAROLINA, DEPARTMENT OF BIOLOGICAL SCIENCES

Oct 2022 – present

Columbia, SC (remote)

- Evolutionary phylogenomic analyses to understand the history of adaptive introgression in *Penstemon* section *Habroanthus*.
- Implementation of predictive models relating spectral and pigmentation phenotypes in natural *Penstemon* hybrids.
- Population genetics and genomics. Analysis of local ancestry and admixture mapping to identify genes and SNP markers associated with the adaptive hummingbird syndrome in *Penstemon*.
- Transcriptomic, biochemical, and morphological characterization of floral development in *Penstemon* hybrids.
- Coordination of project to characterize local metabolic adaptation along an elevation gradient in *Penstemon virens*.
- Field work to collect *Penstemon* vouchers, tissue samples, floral phenotype measurements, and pollinator visitation data in the Rocky Mountains and Southwest Desert.

Postdoctoral Associate

UNIVERSITY OF COLORADO, DEPARTMENT OF ECOLOGY AND EVOLUTIONARY BIOLOGY

Jan 2018 – Apr 2022

Boulder, CO

- Coordination of field trips in Brazil and Argentina to collect samples from the Petunieae clade of Solanaceae. Phylogenetics to reconstruct species relationships. Transcriptomics, biochemical analyses, and bioinformatics to study the evolution of floral traits. Handling large datasets and corresponding metadata.
- Development of computational approaches to 1) simulate the evolution of metabolic pathways and 2) perform segregant analyses of back-cross transcriptomic data to identify genes responsible for flower color transitions.
- Acquisition of funding for the development of computational tools, management of data storage and organization for large-scale international collaborative projects.
- Volunteer efforts: Field work at Niwot Ridge Long-term Ecological Research Program collecting soil moisture and plant community composition data, and in the desert Southwest collecting samples of *Physalis* to study self-incompatibility and *Larrea* to study polyploidization.

Graduate Research Fellow

UNIVERSITY OF OREGON, DEPARTMENT OF CHEMISTRY AND BIOCHEMISTRY

Sep 2012 – Dec 2017

Eugene, OR

- Development of high-throughput binding experiments and implementation of statistical and machine learning approaches to quantitatively measure the evolution of protein-peptide interaction specificity in the S100 protein family.
- Phylogenetics and biochemical analyses to characterize the evolution of metal binding affinity, specificity, and structural effects in the S100 protein family.
- Managed laboratory tasks, supply ordering, sample storage information. As first student, helped to build the lab infrastructure and foundational projects.

Undergraduate Research Assistant

MONTANA STATE UNIVERSITY, DEPARTMENT OF CHEMISTRY AND BIOCHEMISTRY

Sep 2009 – June 2012 (20 hr/week)

Bozeman, MT

- Engineering of virus-like particles as templates for biomedical applications, including analgesics and anti-fungal agents.
- Fluorescent labeling of biomimetic nano-materials for *in vivo* experiments in mice and fungi.

Volunteer

COLORADO NATIVE PLANT SOCIETY

2022 – present

Colorado

- Participant in the Adopt-a-Rare-Plant program: Re-surveying historical collection sites for rare and imperiled endemic plants in Colorado to assess populations in difficult-to-reach locations.
- Seminar speaker: Providing educational talks for the public on Colorado native *Penstemons* and the evolution of their close relatives in neighboring states.

Volunteer Member/Vice-chair/Chair

2021 – present

COLORADO DEPARTMENT OF AGRICULTURE NOXIOUS WEED ADVISORY COMMITTEE

Colorado

- At-large voting member of the committee and vice-chair: advising the CDA Noxious Weed Program on invasive-species management across the state of Colorado.
- Revising the noxious weed assessment process via incorporation of scientific information and the consideration of the needs of various stakeholders across Colorado.
- Member of the Noxious Weed grant committee, helping to decide how CDA funding is distributed to counties, municipalities, and organizations around the state.

Volunteer worker

2005 – 2018

E.H. McCLEERY BUFFALO WOLF FOUNDATION

Emigrant and Bridger, MT

- A non-profit organization formerly operated by my family. Helped to manage and carry out various projects including construction, facility and equipment maintenance, animal vaccination, feeding, and relocation, acquisition and transport of supplies, website administration, photography, relocation of facilities and transfer of foundation ownership to Wolf Haven International.

Publications

*Corresponding author †Undergraduate co-author ‡Co-first author

Pezzi, PH, **Wheeler LC**, Brandão de Freitas L, Smith SD (2024). [Incomplete lineage sorting and hybridization as drivers of tree discordance in *Petunia* and related genera \(Petunieae, Solanaceae\)](#). *Molecular Phylogenetics and Evolution*

Stevens JTE, **Wheeler LC**, Williams NH, Norton AM, Wessinger CA (2023). [Predictive links between petal color and pigment quantities in natural *Penstemon* hybrids](#). *Integrative and Comparative Biology*

Wheeler LC*, Dunbar-Wallis A, Schutz K, Smith SD (2023). [Evolutionary walks through flower color space driven by gene expression in *Petunia* and allies \(Petunieae\)](#). *Proceedings of the Royal Society B: Biological Sciences*

Wheeler LC*†, Walker JF, Ng J, Deanna R, Dunbar-Wallis A, Backes A, Pezzi PH, Virginia Palchetti M, Roberston HM, Monaghan A, Brandão de Freitas L, Barboza GE, Moyroud E, Smith SD (2022). [Transcription factors evolve faster than their structural gene targets in the flavonoid pigment pathway](#). *Molecular Biology and Evolution*

Sinnott-Armstrong M, Deanna R, Pretz C, Liu S, Harris J Dunbar-Wallis A, Smith SD, **Wheeler LC*** (2022). [How to approach the study of syndromes in macroevolution and ecology](#). *Ecology and Evolution*

Huffine CA, **Wheeler LC**, Wing BA, Cameron JC (2021). [Computational modeling and evolutionary implications of biochemical reactions in bacterial microcompartments](#). *Current Opinion in Microbiology*

Wheeler LC, Harms MJ (2021). [Were ancestral proteins less specific?](#) *Molecular Biology and Evolution*

Wheeler LC*, Wing BA, Smith SD (2020). [Structure and contingency determine mutational hotspots for flower color evolution](#). *Evolution Letters*

Wheeler LC*†, Perkins A, Wong CE†, Harms MJ (2020). [Learning peptide recognition rules for a low-specificity protein](#). *Protein Science*

Wheeler LC*, Smith SD (2019). [Computational modeling of anthocyanin pathway evolution: Biases, hotspots, and trade-offs](#). *Integrative and Comparative Biology*

Hiranmayi Duvvuri†, **Wheeler LC**, Harms MJ (2018). [pytc: open source python software for global analyses of isothermal titration calorimetry data](#). *Biochemistry*

Wheeler LC, Anderson JA, Morrison AJ, Wong CE†, Harms MJ (2017) [Conservation of specificity in two low-specificity proteins](#). *Biochemistry*

Wheeler LC, Harms MJ (2017). [S100A5 binds Ca²⁺ and Cu²⁺ independently](#). *BMC Biophysics*

Wheeler LC, Donor MT, Prell JS, Harms MJ (2016). [Multiple evolutionary origins of ubiquitous Cu²⁺ and Zn²⁺ binding in the S100 protein family](#). *PLoS ONE*

Wheeler LC*†, An-Lim S, Marqusee S, Harms MJ (2016). [The thermostability and specificity of ancient proteins](#). *Current Opinions in Structural Biology*

Strengths & Skills

Data management	Organization of large datasets and collection metadata, automated data backup, Git/Github
Project coordination	Management of collaborative projects, organization of symposia and seminars
Interpersonal skills	Collaboration on large projects, working in teams, mentoring students, serving on committees, teaching
Communication & Scholarship	Writing research publications and detailed literature reviews, public and academic presentations
Botanical Field Work	Plant ID & collection, tissue preservation, reflectance spectroscopy, photography, microscopy, cytogenetics
General Field Skills	Navigation (maps, GPS), manual vehicle operation, hiking/backpacking, trip planning, sample transport
Computational specialties	Molecular phylogenetics & evolution, biostatistics, transcriptome assembly, gene expression, simulations
Molecular Biology	PCR, cloning, RNA & DNA extraction, gel electrophoresis, NGS library prep, bacterial culture, phage display
Biochemical techniques	Protein expression & purification, isothermal titration calorimetry, CD spectroscopy, HPLC, ultracentrifugation
Programming languages	Python (> 5 yrs), R (> 5 yrs), Bash (> 5 yrs), Julia (2 yrs), Snakemake (2 yrs), C++ (< 1 yr), SQL (< 1 yr)
Spoken Languages	English (native speaker), Spanish (conversational, proficient in reading and writing)), French (intermediate)

Funding

2019	Research Innovation Office Seed Grant , \$50,000	CU Boulder
2015	Mechanisms of Protein Evolution III: Student Travel Award , \$600	SMBE
2013-2016	Molecular Biology and Biophysics Training Grant (UO-IMB) , \$75,000	NIH
2011-2012	Montana INBRE Research Award , \$1500	MT INBRE
2011	Swager Summer Research Award , \$600	MSU
2011	Geer-Howald-Callis Summer Research Award , \$1200	MSU

Workshops, trainings, & courses

2021	Learn C++ course , Codecademy	Online
2021	Learn SQL course , Codecademy	Online
2021	Diversity, Equity, and Inclusion Workshop , CU	Boulder, CO
2020	ALTEC Spanish Advanced Conversation , CU	Boulder, CO
2018	Navigating the NSF Review Process as a Beginning PI , CU	Boulder, CO
2017	Individual Development Plan (IDP) workshop , UO	Eugene, OR
2016	Illumina Metagenomics Workshop , UO	Eugene, OR

Teaching Experience

Guest lecturer HERPETOLOGY Invited to give lecture and lead discussion on rattlesnake venom evolution	University of Colorado (Boulder) 2022
Guest lecturer COMPARATIVE GENOMICS Invited to give lecture and lead discussion of the role of structural variants in adaptation and speciation	University of Colorado (Boulder) 2021
Guest lecturer PHYLOGENETICS AND COMPARATIVE BIOLOGY Invited to give lecture and lead discussion of forensic phylogenetics for tracing the origin of viral infections	University of Colorado (Boulder) 2021
Co-instructor PLANTS AND SOCIETY Assembled remote lectures and assignments on topics ranging from genetic engineering to ethnobotany	University of Colorado (Boulder) 2020
Guest lecturer CONVERGENCE AND CONSTRAINT: LESSONS FROM THE EVOLUTION OF FLOWER COLOR Invited to give a remote guest lecture on pleiotropy and genetic constraints for upper level evolution class	University of Hawai'i at Mānoa 2020
Guest Instructor BIOCHEMISTRY OF FLOWER COLOR Developed and taught a lecture and lab exercise for visiting high-school students with CU Upward Bound	University of Colorado (Boulder) 2018
Instructor BIOCHEMISTRY RECITATION: METABOLISM Led article discussions and group activities for this course to supplement the primary lecture course	University of Oregon (Eugene) 2013
Teaching assistant BIOCHEMISTRY LABORATORY Helped to plan, organize, and lead the laboratory that accompanied the upper-level biochemistry course	University of Oregon (Eugene) 2013
Teaching assistant GENERAL CHEMISTRY LABORATORY Helped to test activities and lead the laboratory that accompanied the introductory chemistry course	University of Oregon (Eugene) 2012
Volunteer Instructor ADVANCED 5TH GRADE MATH Designed and taught a pre-algebra course for a group of advanced 5th grade students	Juniper Elementary (Bend, OR) 2007

Mentorship & Service

RESEARCH MENTORSHIP

Lily Elliott UNDERGRADUATE STUDENT Mentored a project to study local metabolic adaptation in <i>Penstemon</i>	Boulder, CO 2023-2024
Clair Huffine PHD STUDENT Co-mentored a project to model the cyanobacterial carboxysome	Boulder, CO 2021-2022
Mikayah Oxendine UNDERGRADUATE (SUMMER STUDENT) Oversaw mentorship (by graduate student Sukuan Liu) of project on carnivorous plants in Southeastern North America	Remote 2020
Chandra Jacobs UNDERGRADUATE (SUMMER STUDENT) Oversaw mentorship (by graduate student Jesse Harris) of project on phylogenetic relationships between medicinal plants	Remote 2020
Chantelle Yazzi UNDERGRADUATE (SUMMER STUDENT) Oversaw mentorship (by graduate student Chelsea Pretz) of project on analysis of flavonols in Navajo Tea (<i>Thelesperma</i>)	Remote 2020
Ashley Hamilton UNDERGRADUATE (SUMMER REU STUDENT) Guided a project in the bioinformatic analysis of transcriptomic data from the genus <i>Lochroma</i>	Boulder, CO 2019
Alice Backes VISITING MASTERS STUDENT Guided a project in the bioinformatic analysis of transcriptomic and genomic data in online databases	Boulder, CO 2018
Jocelyn Hernandez UNDERGRADUATE (SUMMER PROGRAM FOR UNDERGRADUATE RESEARCH) Helped student to develop tools to study epistasis in protein evolution	Eugene, OR 2017
Amber Rolland PHD ROTATION STUDENT Helped to guide a project studying metal binding in the S100A8/A9 heterodimer complex	Eugene, OR 2016
Erik Burlingame MASTERS STUDENT Guided a group project to develop an analysis pipeline for high-throughput phage display sequencing data	Eugene, OR 2016
Samantha Sivagnanam MASTERS STUDENT Guided a group project to develop an analysis pipeline for high-throughput phage display sequencing data	Eugene, OR 2016
Wendy Su MASTERS STUDENT Guided a group project to develop an analysis pipeline for high-throughput phage display sequencing data	Eugene, OR 2016
Caitlyn Wong UNDERGRADUATE (HONORS THESIS) Guided a project characterizing the structural response of S100 proteins to binding of metals and peptides	Eugene, OR 2015 - 2017
Hiranmayi Duvvuri UNDERGRADUATE STUDENT Helped student to learn techniques and carry out isothermal titration calorimetry experiments	Eugene, OR 2015-2016
Sarina Pollat UNDERGRADUATE (SUMMER PROGRAM FOR UNDERGRADUATE RESEARCH) Guided a project measuring the thermodynamics of metal and peptide binding in S100 proteins	Eugene, OR 2015
Abigail Tami UNDERGRADUATE (SUMMER PROGRAM FOR UNDERGRADUATE RESEARCH) Guided a project measuring the thermodynamics of metal and peptide binding in S100 proteins	Eugene, OR 2014
Sunny Ketchum GRADUATE ROTATION STUDENT Guided a project developing the molecular tools to study the S100A8/A9 protein complex	Eugene, OR 2014
Kevin McKnaught GRADUATE ROTATION STUDENT Helped student to develop a method for expression and purification of certain S100 proteins	Eugene, OR 2014
Kendall Saboda UNDERGRADUATE STUDENT Helped student with the engineering and molecular cloning of custom virus-like particles	Bozeman, MT 2012
Kristen Hyberg UNDERGRADUATE STUDENT Helped student with the molecular cloning, expression, and purification of custom virus-like particles	Bozeman, MT 2011-2012
SERVICE ACTIVITIES	
International Botanical Congress CO-ORGANIZER Symposium - Solanaceae: Biology, Systematics, and Evolution.	Spain 2024
Solanaceae seminars CO-ORGANIZER Helping to organize the SOL Seminar Online international seminar series	Colorado 2021-present
CDA Noxious Weed Advisory Committee MEMBER/VICE-CHAIR At-large member, helping to advise CDA on strategies for noxious weed management and organize committee activities	Colorado 2021-present

Colorado Native Plant Society MEMBER Providing educational presentations, leading field trips, participating in rare plant surveys	Colorado 2022-present
CCL Colorado Virtual Seminars CO-ORGANIZER Helped to organize interdisciplinary seminar and discussion group on climate policy	Colorado 2020-2022
Smith lab Summer internship for Native American undergraduates VOLUNTEER Helped to coordinate a Summer research experience for former students from the CU Upward Bound program	Remote/in-person 2020-2021
Bridge Program Choose Research Panel VOLUNTEER Sat on a panel discussion for community college transfer students interested in undergraduate research	Remote 2020
McKenzie school district outreach VOLUNTEER Science discussions for 2nd grade classroom in the McKenzie school district in Oregon	Remote 2019-present
CU Upward Bound INSTRUCTOR Leading virtual and in-person outreach activities for high school students	Colorado 2018-present
Pinhead Institute INSTRUCTOR Leading "Punk Science" outreach activities for K-12 students	Telluride, CO 2019
2019 Mayors' Climate Panel CO-ORGANIZER Helped to organize the 2019 Mayors' Climate Panel. Managed online registration and speaker invitations	Denver, CO 2019
Citizens' Climate Lobby VOLUNTEER Helped to organize tabling events, educational presentations, and lobby meetings with US representatives	Golden, CO 2018 - present
Quantitative Problem Solving & Research Communication Consortium CO-CHAIR & CO-FOUNDER Helped to organize presentations & brainstorming sessions for graduate students and postdocs	Eugene, OR 2016 - 2017
UO SafeRide Program VOLUNTEER Helped to provide free, safe rides at night for members of the university community	Eugene, OR 2016 - 2017
Mad Duck Science Fridays VOLUNTEER Helped to organize and run engaging science learning activities for middle school students	Eugene, OR 2013 - 2017
Associated Students of MSU STUDENT SENATOR Served on several committees and acted as liaison to several student groups	Bozeman, MT 2010 - 2011
Undergraduate Chemistry Society SECRETARY & VICE-PRESIDENT Helped to organize fundraisers and to provide educational activities for local and rural Montana schools	Bozeman, MT 2009 - 2012
Sacajawea Middle School VOLUNTEER MENTOR Mentored a gifted student in advanced math; oversaw development of a project on fractals	Bozeman, MT 2009-2010
Juniper Elementary School VOLUNTEER MENTOR Provided tailored educational activities to students with special needs	Bend, OR 2005 - 2008

Presentations and Seminars

Sol International Conference "PHYLOTRANSCRIPTOMIC ANALYSIS OF FLOWER COLOR EVOLUTION IN PETUNIEAE" • Submitted poster	Online November 2020
Solanaceae Seminar Series "AN INTEGRATIVE APPROACH TO STUDYING THE EVOLUTION OF FLORAL PIGMENTATION" • Invited talk	Online September 2020
Botany virtual meeting "PHYLOTRANSCRIPTOMICS OF THE PETUNIEAE CLADE OF SOLANACEAE" • Submitted talk	Online July 2020
Evolution meeting "PREFERENTIAL FIXATION OF STRUCTURAL OR REGULATORY MUTATIONS DEPENDS ON PATHWAY POSITION" • Submitted talk	Providence, RI June 2019
Society for Integrative and Comparative Biology "COMPUTATIONAL MODELING OF ANTHOCYANIN PATHWAY EVOLUTION" • Submitted talk	Tampa, FL Jan. 2019
Geobiology Supergroup Seminar "COMPUTATIONAL MODELING OF FLOWER COLOR EVOLUTION" • Invited talk	Boulder, CO Nov. 2018

Quantitative Think Tank

“DEVELOPING A COMPUTATIONAL MODEL OF THE ANTHOCYANIN PATHWAY”

- Submitted talk

Society for Molecular Biology and Evolution

“PHAGE DISPLAY AND DEEP SEQUENCING TO STUDY THE EVOLUTION OF BINDING SPECIFICITY”

- Submitted poster

Institute of Molecular Biology Seminar Series

“EVOLUTION OF METAL AND PEPTIDE BINDING IN THE S100 PROTEIN FAMILY”

- Annual seminar

Third International Symposium on Protein Folding and Dynamics

“TRACING THE EVOLUTION OF PEPTIDE BINDING SPECIFICITY IN THE S100 PROTEIN FAMILY USING PHAGE DISPLAY AND DEEP SEQUENCING”

- Submitted talk

Gibbs Conference on Biological Thermodynamics

“PREFERENTIAL FIXATION OF STRUCTURAL OR REGULATORY MUTATIONS DEPENDS ON PATHWAY POSITION”

- Submitted talk

Life at the Nanoscale mini-symposium

“TRACING THE EVOLUTION OF PEPTIDE BINDING SPECIFICITY IN THE S100 PROTEIN FAMILY USING PHAGE DISPLAY AND DEEP SEQUENCING”

- Submitted poster

Institute of Molecular Biology Seminar Series

“TRACING THE EVOLUTIONARY HISTORY OF TRANSITION-METAL BINDING AND PEPTIDE BINDING SPECIFICITY IN THE S100 PROTEIN FAMILY”

- Annual seminar

Mechanisms of Protein Evolution III: Origins

“TRACING THE EVOLUTIONARY FLUCTUATIONS OF PEPTIDE BINDING SPECIFICITY IN THE S100 PROTEIN FAMILY”

- Submitted talk

Gibbs Conference on Biological Thermodynamics

“TRACING THE EVOLUTIONARY FLUCTUATIONS OF PEPTIDE BINDING SPECIFICITY IN THE S100 PROTEIN FAMILY”

- Submitted poster

Protein Folding Consortium Workshop

“PROBING THE EVOLUTIONARY HISTORY OF PEPTIDE BINDING SPECIFICITY IN THE S100 PROTEIN FAMILY”

- Submitted talk

Institute of Molecular Biology Seminar Series

“PROBING THE EVOLUTION OF PEPTIDE-BINDING SPECIFICITY IN THE S100 PROTEIN FAMILY”

- Annual seminar

Gibbs Conference on Biological Thermodynamics

“EVOLUTIONARY BIOPHYSICAL STUDIES OF PEPTIDE SPECIFICITY IN THE S100S”

- Submitted poster

Grad Talk Series

“EVOLUTION AS A MOLECULAR MATCH MAKER: HOW DO BIOLOGICAL MOLECULES EVOLVE TO RECOGNIZE EACH OTHER?”

- Invited talk

Protein Folding Consortium Workshop

“EVOLUTIONARY BIOPHYSICAL STUDIES OF PROTEIN FUNCTION IN THE S100 FAMILY”

- Submitted poster

Boulder, CO

Oct. 2018

Austin, TX

July 2017

Eugene, OR

May. 2017

Bangalore, India

Nov. 2016

Carbondale, IL

Sep. 2016

Eugene, OR

Jun. 2016

Eugene, OR

Apr. 2016

Denver, CO

Nov. 2015

Carbondale, IL

Sep. 2015

Berkeley, CA

May. 2015

Eugene, OR

May. 2015

Carbondale, IL

Sep. 2014

Eugene, OR

Nov. 2014

Ann Arbor, MI

May. 2014